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Title- Part 1 Report

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Long Question

1. You are a Data Scientist at a leading digital payment platform such as eSewa, working on large-scale user transaction data.

- Identify two high-impact business problems where unsupervised learning can provide value (e.g., customer segmentation, fraud detection, behavioral profiling).

- For each problem:

- Clearly define the problem in a real-world context.

- Propose a suitable unsupervised learning approach (e.g., clustering, anomaly detection, dimensionality reduction) and recommend specific algorithms (e.g., K-Means, DBSCAN, Autoencoders).

- Discuss how the model outputs can be integrated into real-time or batch systems to support business decisions.

- Briefly mention one limitation or challenge of your proposed approach.

- Assume limited labeled data and large-scale transactions.

Answer

-Problem 1: Real-Time Transactions Fraud Detection

Understanding the problem: eSewa handles thousands of payments every day and frauds (unusual payment behavior, user account takeover, money laundering) lead to substantial revenue loss and a loss of trust. We need to detect fraud transactions automatically, but have few labeled fraud instances.

Proposed Approach: Use anomaly detection with Autoencoders. Isolation Forest quickly identifies anomalies by randomly segmenting data, and Autoencoders learn the normal patterns of transactions and identify anomalies by the reconstruction error.

Integration into Systems:

- Real-time: Run the trained model as a microservice to score transactions in real-time. Alerts are triggered, or additional verification (OTP, human review) is required for transactions with an anomaly score exceeding a threshold.
- Batch: Schedule daily batch processes to process historical transactions, detect new fraud patterns and update the model with fresh data.
- Challenge: False positives may annoy legitimate users with extra verification, needing to set thresholds and involve humans in verification to avoid user drop-off.

-Problem 2: Segmentation for targeted Marketing

Problem Statement: eSewa has millions of customers with different transaction patterns (volume, value, merchant categories, time of transactions). Lacking customer segmentation, marketing strategies are generic and the response rate low, resulting in poor engagement and conversion.

Proposed Approach: Clustering algorithms (K-Means or DBSCAN). K-Means clusters users into a fixed number of segments (e.g., high-value, occasional or dormant) using features such as number of transactions, average purchase, merchant type and recency. DBSCAN will find oddly-shaped clusters and outliers, without needing to know how many clusters there are.

Integration into Systems:

- Batch: Cluster users' transactions weekly/monthly. Label user with segment and store in the database.
- Real-time: Use segment tags to send personalized push notifications (cashback for valued users, re-activation for inactive users) via email and/or mobile app. Business uses segment labels for targeted marketing, product recommendations and loyalty schemes.
- Limitation: Clusters may not have immediate business meaning, and number of clusters (for K-Means) needs to be experimented with. Users' purchasing patterns change, so we need to re-train the model periodically to update segments.

Short Question

Overfitting

1. In the context of machine learning:

- Define overfitting and underfitting, and clearly distinguish between the two in terms of model behavior on training and unseen data.
- Explain why both overfitting and underfitting negatively impact generalization performance.
- Illustrate your explanation using simple examples (e.g., a model memorizing training data vs failing to capture complex patterns in the data).

Answer

- Overfitting occurs when the model becomes excessively learned on the training data, including noise and minor details. It works well on training data, but not on unknown data.

Underfitting occurs when a model is so simplistic that it is not able to learn the data pattern. It does not do well on training as well as unseen data.

Generalization means how well a model works on new, unseen data.

- In overfitting, the model learns nothing but remembers the patterns thus failing on new data.
- In underfitting, the model fails to learn, thus failing to perform well even on training data. The two cases result in poor accuracy with real-world data.

Simple examples:

- **Overfitting example:** The model learns to produce answers to old exam questions but is unable to answer new questions
- **Underfitting example:** A model is a straight line that fits complicated curved data, omitting significant patterns.

2. Overfitting is a common challenge in deep learning, especially when models have high capacity.

- Describe at least two techniques used to reduce overfitting (e.g., dropout, early stopping, data augmentation, L2 regularization).

- For each technique:

- Explain the intuition behind how it works.

- Describe how it affects the training process or model behavior.

- Provide a real-world application example (e.g., image classification, text classification, fraud detection).

Answer

-Methods to minimize Overfitting:

i. Data Augmentation

-The principle is to add more training data and variety by simple modifications to existing data. As an illustration, pictures may be turned, flipped or even altered slightly. This allows the model to recognize numerous variations of the same input hence it does not memorize particular examples; it acquires the general patterns. This is essentially because during training, the data is essentially larger and more diverse, thereby enhancing the ⁴ model to perform well on new unseen data. This has a frequent application in the real world in image classification, e.g. in medical imaging, where a small amount of data can be extended using transformations.

-Effect on training:

- More diverse data
- Model does not learn by memorization, but learns general features.

-Real-world example:

- Image recognition (e.g. medical scans)

It is possible to rotate or zoom same image to generate additional training samples.

ii. ¹ Early Stopping

-Early stopping involves monitoring the performance of the model on validation data and halts training when the performance begins to deteriorate. It is the intuition that at some stage, the model will start memorizing noise rather than picking up some useful patterns. This has implications on training whereby it eliminates unnecessary epochs and overfitting. It is also time saving and computationally saving. An example is the text classification, e.g. spam, where training is terminated at a point where no further accuracy increases on new data.

-Effect on training:

- Validation error begins to increase: this terminates training.
- Prevents over-training

-Real-world example:

- Text (e.g. spam) classification.
Training stops once performance on unseen data stops improving.

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